

AMNA MUBARAK

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EDUCATION

BS Software Engineering

FAST National University of Computer and Emerging Sciences

Sep 2020 – June 2024

Karachi, Pakistan

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

Backend: FastAPI, Django, Django Ninja, Flask, REST APIs, Microsoft Graph API

Databases: PostgreSQL, Azure SQL Server, MongoDB, Pinecone (Vector DB), Oracle, AWS Redshift

AI / ML: RAG Systems, LangChain, OpenAI API (GPT-4), Prophet, Holt-Winters, PyTorch, Scikit-Learn

Data Engineering: ETL Pipelines, Time-Series Forecasting, Schema Design, Matillion, Data Validation

Cloud & DevOps: AWS (Redshift, S3, EC2), Azure (SQL, Blob Storage, Key Vault), Docker, GitHub Actions

Testing: PyTest, Selenium, Playwright

EXPERIENCE

Blue Ocean Strategic Partners

Dec 2024 – Present

Software Engineer II

Remote, US-based

- Own end-to-end automation and backend systems, collaborating directly with clients to scope requirements and deliver production-ready solutions in a fast-paced startup.
- Leading development of an ESG analytics platform processing **53K+ utility records** across **430+ hotel properties** with time-series forecasting (Holt-Winters, Prophet) for gap-filling and reporting.
- Replaced legacy Azure Logic Apps with a Python + Microsoft Graph API service with rate limiting and pagination, automating **2,000+ weekly financial reports** and **reducing execution time by 60%**.
- Designed scalable retry and monitoring mechanisms to handle API failures and data inconsistencies, ensuring **zero manual intervention** in recurring workflows.

Software Engineer I

- Automated Oracle payment processing pipeline handling **5,000+ monthly transactions**, eliminating manual reconciliation and reducing financial close time through modular validation and retry logic.
- Developed an OCR-based document parser to automatically detect and split multi-page invoices for Hilton and Marriott with **99% accuracy**, eliminating hours of manual file processing.

Share Mobility

Jan 2024 – Dec 2024

Associate Software Engineer

Remote, US-based

- Built async FastAPI backend for AI ride-booking chat handling **1,000+ daily requests** with GPT-4 intent classification.
- Developed voice-enabled booking assistant using Vapi and GPT-4 with guardrails for safe conversational flows.
- Built semantic search using Pinecone + OpenAI embeddings, **improving support accuracy by 35%**.
- Optimized AWS Redshift ETL pipelines through schema redesign, **reducing query time by 40%**.

PROJECTS

FinAI – AI Finance Assistant | Python, FastAPI, LangChain, Pinecone, PostgreSQL, Docker

[GitHub](#)

- Built RAG assistant parsing 100+ page bank statements into PostgreSQL for structured financial analysis.
- Implemented hybrid retrieval with vector embeddings + metadata filtering for **sub-second queries** across transactions.

Codebase Query Engine | Python, FastAPI, OpenAI, Pinecone, Docker

[GitHub](#)

- Built RAG developer assistant indexing source code with syntax-aware chunking for semantic code search.
- Engineered async FastAPI backend for real-time repository ingestion and vector retrieval, containerized with Docker.

PredictFlow – ML Prediction Service | Python, FastAPI, Docker, Celery, Redis

[GitHub](#)

- Built scalable ML prediction service with sync and async processing using Celery workers and Redis message broker.
- Designed modular architecture with pluggable model backends, health checks, and Docker-based deployment.